SOFTWARE ENGINEERING MINI PROJECT · TEAM E15

Test Plan

Railway Reservation System

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PES1UG24CS276

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ISO/IEC/IEEE 29119-3-aligned structure

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Document Control

Document	Test Plan
System	Railway Reservation System (RRS)
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Revision History

Version	Date	Change	Author
1.0	29 Sep 2026	Initial Part 1 plan and ten case specifications.	Atharv More / Muhammad Affan Dumba
1.1	29 Sep 2026	Added explicit password-boundary/hash, late-cancellation and booking/cancellation-audit checks; separated submission copy.	Team E15 review

Execution status. The cases in §5.2 are specifications, not claims of execution. Actual result, evidence, tester, date, build and status are recorded when an implementation is available.

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1. Introduction

1.1 Purpose

This plan defines the scope, strategy, resources, environment, entry/exit criteria, security validation and traceability for system testing of RRS. The SRS v1.1 is the requirements baseline.

1.2 Objectives

- Verify all functional, non-functional and security requirements.
- Verify booking/cancellation invariants under normal, failure and concurrent conditions.
- Produce repeatable evidence and a requirement-level completion review.

1.3 References

Team E15 RRS SRS v1.1; Team E15 Architecture and Design Specification v1.1; ISO/IEC/IEEE 29119-3:2021 test documentation guidance.

2. Test Items and Scope

In scope	Registration; authentication; search; availability; booking; history; cancellation; administration; audit; performance; concurrency; restart recovery; accessibility; compatibility; transport, session, credential and authorization security.
Test items	Browser UI, REST API, domain services, relational database behavior and payment-simulator integration.
Out of scope	Real card payments, live railway feeds, production disaster recovery, dynamic pricing and wait-list allocation.
Assumptions	Seeded synthetic data, test-only database inspection, configurable payment outcomes and current supported browsers are available.

3. Test Strategy and Approach

3.1 Test Levels

- **Unit:** validation, password policy, state transitions and availability calculations.
- **API integration:** authorization, database constraints, persistence, idempotency and simulator outcomes.
- **System:** passenger/administrator browser workflows, accessibility and compatibility.
- **Non-functional:** load, concurrency and restart recovery.

3.2 Techniques

Positive/negative scenarios, equivalence partitions, boundary values, role/ownership checks, duplicate submission, state-transition testing and concurrent final-seat requests are used. Automated checks are preferred for API, load and security work; keyboard and visual compatibility checks may be manual with recorded evidence.

3.3 Test Data

Use synthetic passenger accounts P1 and P2, administrator A1, trips with zero/one/multiple seats, a departed trip, success/failure simulator modes and a 10,000-trip performance seed. Reset independent cases to a known database snapshot.

3.4 Defect Handling

Record case ID, build, environment, reproducible steps, expected/actual results, severity and evidence. Retest a fix and execute affected regression cases. Critical or high defects block exit unless a documented academic-review decision states otherwise.

4. Environment, Responsibilities and Schedule

4.1 Environment

- HTTPS test deployment and isolated relational test database.
- Current stable Chrome, Edge and Firefox; 375-pixel and 1280-pixel viewports.
- Load/concurrency tool supporting at least 50 virtual users and synchronized requests.
- Authorized, read-only database/log inspection for credential and audit validation.

4.2 Responsibilities

Member	Responsibility
Atharv More	Test Plan ownership, environment coordination, execution tracking and exit review.
Muhammad Affan Dumba	§5.2 test-case specifications, coverage maintenance and evidence checklist.
Mohit Singh P	Requirement mapping review.
Mohammed Rizwan Shaikh	Architecture-related setup, interface and failure-behavior review.

4.3 Sequence

Baseline documents → deploy/seed → smoke test → functional/security cases → performance/ concurrency → accessibility/compatibility → defect retest/regression → traceability and exit review.

4.4 Entry and Exit Criteria

Entry	SRS and design baselined; test build deployed; simulator and seed data available; test accounts and evidence repository prepared.
Exit	All ten cases have recorded outcomes; no open critical/high defects; every FR/NFR/SEC statement is demonstrably covered or has a documented follow-up test; evidence is reviewed.

4.5 Risks and Mitigations

Risk	Mitigation
Environment timing distorts performance	Record hardware/build/tool versions; warm up; repeat runs using the same seed.
30-minute session test is slow	Run the real-duration acceptance check; use configurable time only for earlier regression.
Concurrent requests are not synchronized	Use a barrier so all 20 final-seat requests start together.
Sensitive test data leaks	Use synthetic identities and redact tokens/passwords from evidence.

5. Test Design and Validation

A case passes only when every expected result is observed and retained evidence supports the result. Failure of one assertion fails the case until corrected and rerun.

5.1 Security Validation

ID	Validation	Cases
SV-01	Verify generic authentication failure, fifth-attempt threshold, 15-minute rejection window and password-free audit records.	TC-02
SV-02	Use P2 against P1’s booking and P1 against administrator endpoints; expect 403/404 with no protected data.	TC-06, TC-07
SV-03	Inspect HTTPS, replay a signed-out token and test expiry after 30 minutes of inactivity.	TC-09
SV-04	Inspect authorized test storage/logs: password is a salted adaptive hash and plaintext is absent from database, API responses and logs.	TC-01, TC-02

5.2 Test Cases

TC-01 — Registration, Password Boundary and Credential Storage

Preconditions	No P1 account; authorized test-only read access to user storage and logs.
Procedure	Submit an 11-character password; submit a valid 12-character password and unique email; repeat the same email; inspect stored credential, response and logs.
Expected	11-character input is rejected; valid input creates one account; duplicate is rejected; stored value is a salted adaptive hash and does not equal the password; plaintext appears in neither responses nor logs.
Traceability	FR-01; SEC-02
Evidence	API/UI responses, redacted storage inspection and log search.

TC-02 — Sign-In, Lockout and Authentication Audit

Preconditions	P1 exists and is unlocked.
Procedure	Sign in correctly and sign out; submit five incorrect passwords within 15 minutes; attempt a sixth; inspect audit; retry after the 15-minute rejection period.
Expected	Valid login works; invalid responses are generic; sixth attempt is rejected for 15 minutes; success, failures and rejection are audited without submitted passwords; access resumes after the period with valid credentials.
Traceability	FR-02, FR-09; SEC-02, SEC-04
Evidence	Timestamped responses and redacted audit records.

TC-03 — Search and Availability

Preconditions	Matching, nonmatching, sold-out and multiple-departure trips are seeded.
Procedure	Search by origin, destination and date; inspect filtering, ordering and every availability value.
Expected	Only exact route/date matches appear in departure order; every count is between zero and capacity.
Traceability	FR-03, FR-04
Evidence	Seed extract and ordered API/UI result.

TC-04 — Successful Booking, History and Audit

Preconditions	P1 signed in; selected trip has one free seat; simulator succeeds.
Procedure	Submit one booking; query P1 history and audit records.
Expected	Exactly one confirmed booking with unique PNR and one allocated seat; P1 history shows PNR/trip/seat/fare/status; confirmation audit contains actor, action, time and outcome.
Traceability	FR-05, FR-06, FR-09
Evidence	Booking response, database state, history and audit event.

TC-05 — Payment Failure and Idempotent Retry

Preconditions	P1 signed in; one free seat; known idempotency key.
Procedure	Run with simulator failure; verify state. Then configure success and send the same logical request twice with one new idempotency key.
Expected	Failure confirms nothing and releases the hold; duplicate success calls return one booking/PNR and allocate one seat.
Traceability	FR-05; NFR-03
Evidence	Responses, booking count and inventory before/after.

TC-06 — Ownership, Eligible/Late Cancellation and Audit

Preconditions	P1 owns one future confirmed booking and one departed confirmed booking; P2 signed in separately.
Procedure	P2 reads/cancels P1 future booking; P1 cancels it twice; P1 attempts to cancel the departed booking; inspect inventory/refund/audit.
Expected	P2 gets 403/404 without details; P1’s first eligible cancellation succeeds with one seat release/refund reference/audit event; repeat and late attempts are rejected without further inventory change.
Traceability	FR-06, FR-07, FR-09; SEC-01
Evidence	Responses, inventory deltas, payment event and audit events.

TC-07 — Administrator Inventory, Authorization and Audit

Preconditions	A1 administrator and P1 passenger exist.
Procedure	P1 attempts administrator create; A1 creates, edits and disables a trip; inspect state and audit.
Expected	P1 receives 403; A1 changes persist; each change has actor, timestamp, action and outcome.
Traceability	FR-08, FR-09; SEC-01
Evidence	Responses, final trip state and audit events.

TC-08 — Concurrent Final Seat

Preconditions	Exactly one seat; 20 distinct authenticated passengers; simulator succeeds.
Procedure	Release 20 booking requests simultaneously from a synchronization barrier; inspect bookings and availability.
Expected	At most one confirmation; others report unavailable; final count is zero and is never negative.
Traceability	FR-04, FR-05; NFR-03
Evidence	Tool timestamps, responses, booking count and availability history.

TC-09 — Transport, Session and Restart Recovery

Preconditions	HTTPS deployment; P1 session; committed booking exists.
Procedure	Inspect transport; sign out and replay token; create a session, wait 30 minutes idle and reuse it; restart application; query health and booking.
Expected	Authenticated traffic uses HTTPS; signed-out/expired tokens fail; booking persists; health responds within 60 seconds; logs omit tokens/passwords.
Traceability	NFR-04; SEC-02, SEC-03
Evidence	Transport capture, timestamped responses, restart log and persisted record.

TC-10 — Performance, Accessibility and Compatibility

Preconditions	10,000 trips; 50 virtual users; Chrome/Edge/Firefox; 375px and 1280px viewports.
Procedure	Run 200 searches and 100 bookings at 50 users; complete required workflows by keyboard at both widths in each browser; inspect accessible names.
Expected	≥95% searches ≤2s; ≥95% bookings ≤3s under the simulator condition; keyboard completion succeeds; every required control has an accessible name; no primary-form horizontal scrolling.
Traceability	NFR-01, NFR-02, NFR-05, NFR-06
Evidence	Load summary, browser/version matrix, screenshots and accessibility report.

6. Test Completion and Deliverables

Evidence includes completed case records, API responses/screenshots, performance summaries, accessibility notes, audit/storage inspection, defect log and final traceability review. A failed case is not waived without a recorded decision and reason.

Execution metadata required per case	Build, environment, tester, date/time, actual result, evidence link, pass/fail/block status and defect ID where applicable.
Completion review	Test owner and requirement reviewer confirm evidence and unresolved risks against exit criteria.

7. Requirement Traceability Matrix

Requirement	Planned verification
FR-01	TC-01
FR-02	TC-02
FR-03	TC-03
FR-04	TC-03, TC-08
FR-05	TC-04, TC-05, TC-08
FR-06	TC-04, TC-06
FR-07	TC-06
FR-08	TC-07
FR-09	TC-02, TC-04, TC-06, TC-07
NFR-01	TC-10
NFR-02	TC-10
NFR-03	TC-05, TC-08
NFR-04	TC-09
NFR-05	TC-10
NFR-06	TC-10
SEC-01	TC-06, TC-07
SEC-02	TC-01, TC-02, TC-09
SEC-03	TC-09
SEC-04	TC-02